UNIVERSITY OF PENNSYLVANIA ESE 5700

The 16×4 SRAM macro

Project 1, Part 2: array, decoder, column circuits and timing

Tyrone Marhguy • Victor Wanjohi

Cadence GPDK 18 nm $V_{DD} = 0.8$ V, nominal, 27°C Part 1 cell 0.39 μm^2, $D = 28$ $T_{CLK,\min} = 0.69$ ns
extracted $C_{BL} = 8.59$ fF, $C_{WL} = 4.38$ fF

Team members and contributions

Member	Contribution
Tyrone Marhguy	Part 1 cell layout, Pegasus DRC/LVS, Quantus extraction of `6t_sram_final_design`; tiled 16×4 array `6t_sram_64b`, array DRC, and the C_{BL}/C_{WL} table; report assembly.
Victor Wanjohi	Decoder, column circuits and unit tests; controller; full-macro benches; clock-period, power and access measurements.

Both authors are responsible for the complete design and for every result in this report.

Executive summary

The macro is a 16-word by 4-bit synchronous SRAM of the Part 1 cell. Rows are decoded by a 2-2 predecode (NAND3 enable group, NAND2 address group, NAND2+driver per row). Each column has a 3-PMOS precharge, an 8-fin write driver, a 2:2 inverter on BLB and an output latch. Quantus All-Parasitics on one cell are $C_{BL} = 497.8$ aF, $C_{BLB} = 576.5$ aF and $C_{WL} = 938.9$ aF. The 16-cell column and 4-cell row, plus 0.62 fF of array wire, are the extracted loads used below: $C_{BL} = 8.59$ fF, $C_{BLB} = 9.84$ fF, $C_{WL} = 4.38$ fF. At 0.8 V:

- Write driver: 8 fins, 96 ps at 3.5 fF (102 ps at 4.2 fF). Scaled to the extracted column: 140 ps; at +20% extracted (10.3 fF): 154 ps.
- Read: $t_{BL} = C_{BL}\Delta V / I_{\text{cell}}$. With $I_{\text{cell}} = 25$ μA and $\Delta V = 0.424$ V this is 59 ps at 3.5 fF (measured 83 ps) and 146 ps at 8.59 fF. Design wordline pulse 140 ps, set by the write at the column-test load.
- Decoder: address and WL_EN share the same four-stage path. Predicted 182 ps at 0.5 fF, 260 ps at 4.38 fF. Measured WL_EN→WL 97 ps rising, 101 ps falling, identical at all 16 addresses.
- Controller: CLK→WL 238 ps; thinnest edge margin 70 ps (RD_EN to the next precharge).
- Full macro: $T_{CLK,\min} = 0.69$ ns; read access 488 ps; write time 339 ps; write/read/standby power 124/115/4 μW; area 54.6 μm^2; FOM 4.7×10^3 μm$^2 \cdot \mu$W·ns.

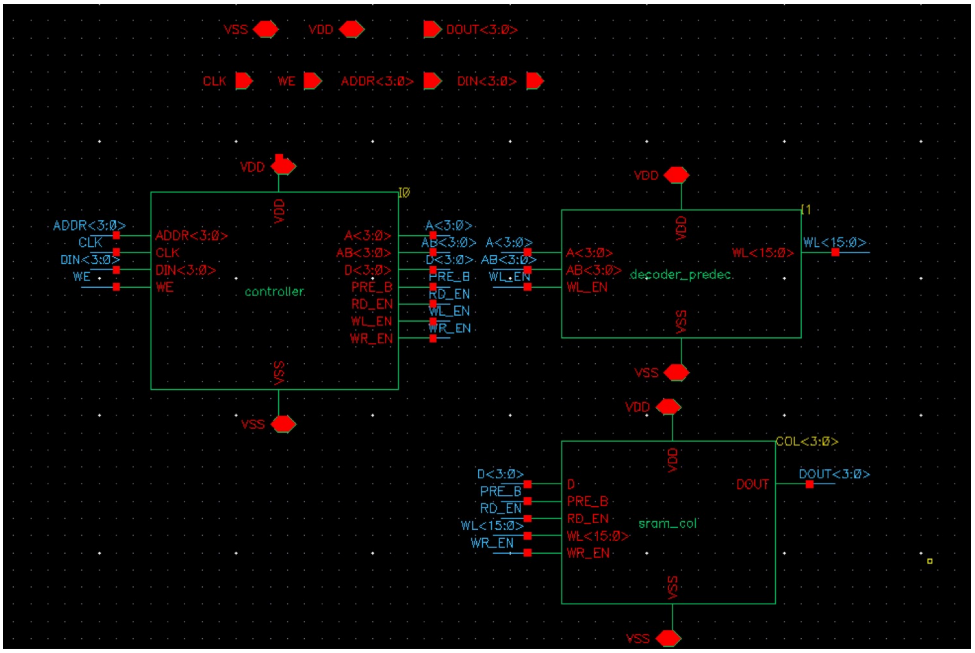


Figure 1: `sram_macro`: controller, `decoder_predec` and four `sram_col` instances. Ports CLK, WE, ADDR⟨3:0⟩, DIN⟨3:0⟩, DOUT⟨3:0⟩ match the handout.

1 Array layout and extraction (2.1)

The array `6t_sram_64b` is 16 rows by 4 columns of the Part 1 cell, mirrored so neighbours share bitline contacts and the VDD/VSS rails. Pitch is the Part 1 PR box: 0.72 μm along the wordline (four columns) and 0.54 μm along the bitline (sixteen rows), so the tile is 2.88×8.64 μm $= 24.9$ μm^2.

Pegasus 23.20 DRC of the array is clean: 0 results in 0 of 374 checks. Pegasus LVS of the tiled cell is ISL clean with comparison match (`6t_sram_final_design`). The array schematic is 64 instances of that cell; abutment joins matching BL, BLB, WL, VDD and VSS pins, so the layout connectivity is the cell LVS plus those shared rails.

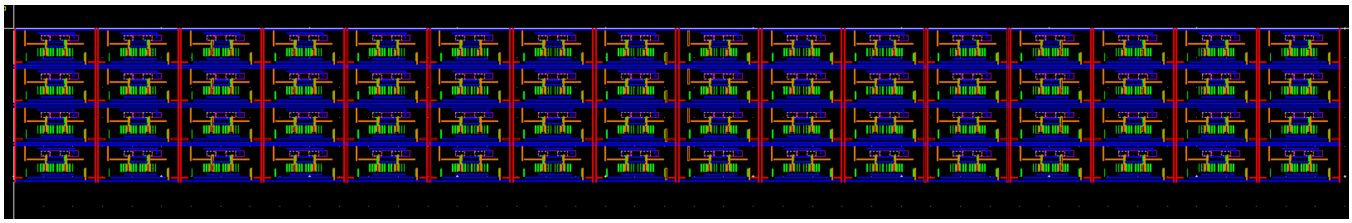


Figure 2: Tiled 16×4 array `6t_sram_64b` (64 bitcells). PR boundary 2.88×8.64 μm.

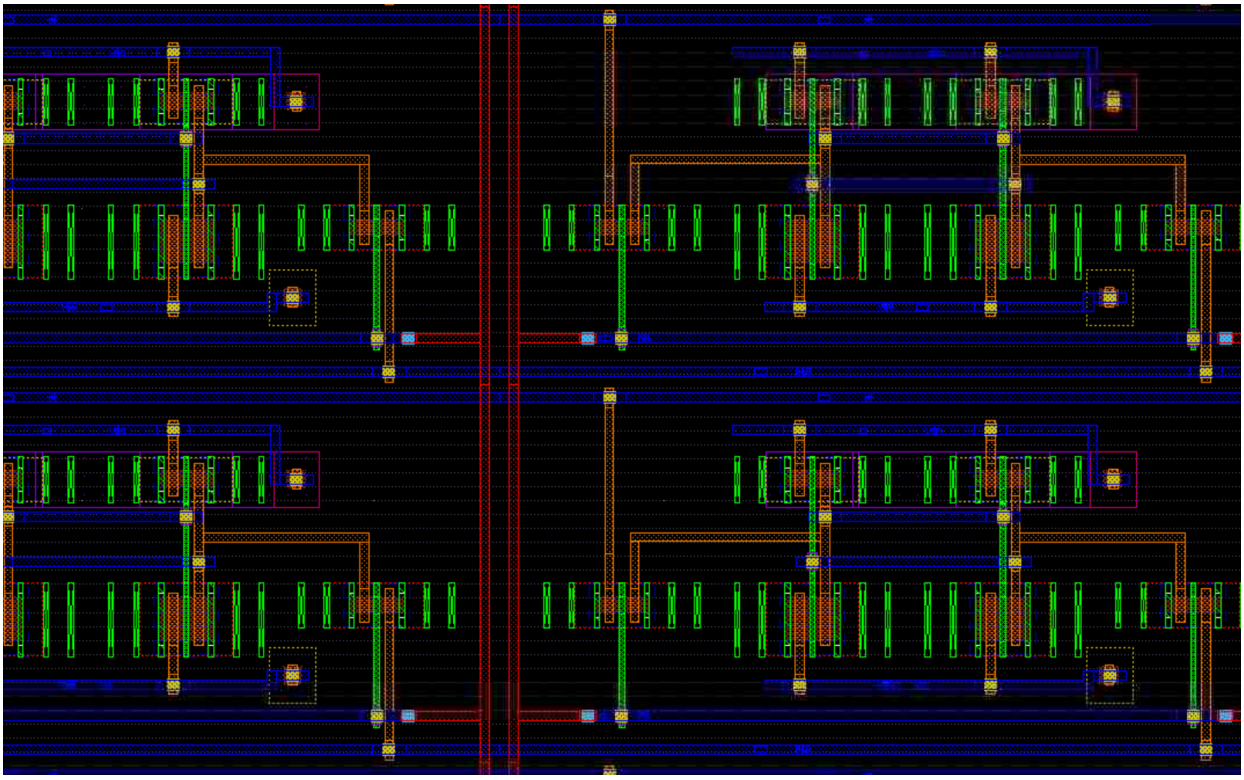
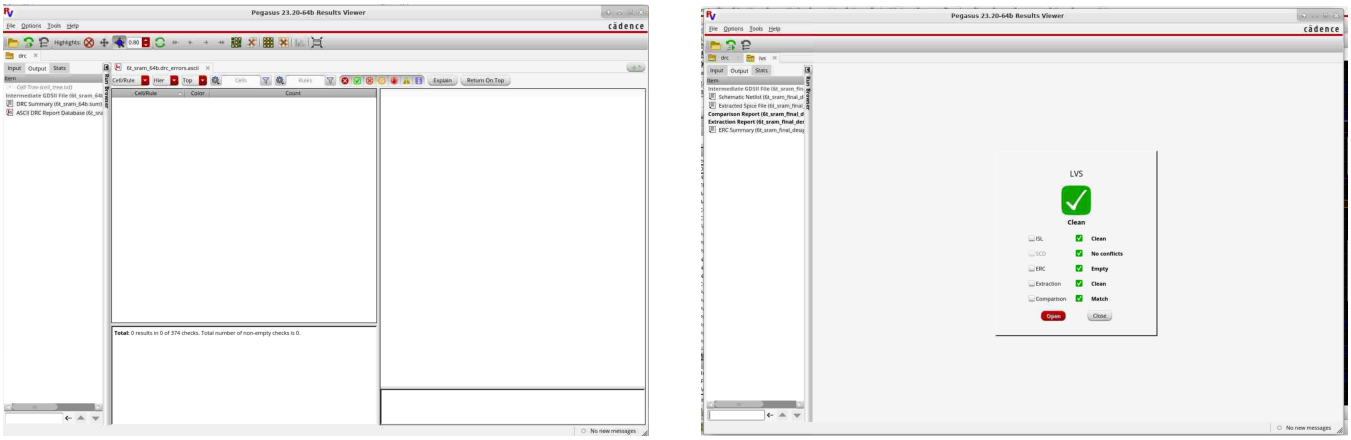


Figure 3: Four mirrored cells sharing the centre bitline contacts and the horizontal supply rails.



(a) Array DRC: 0 of 374. **(b)** Cell LVS: ISL clean.

Figure 4: Pegasus 23.20: DRC of `6t_sram_64b` and LVS of the instance cell `6t_sram_final_design`.

Quantus 23.1.2 on the Part 1 cell (typical, 25°C, `rc_decoupled`) wrote `av_extracted`: 132 R, 97 C, 4 `n1svt` + 2 `p1svt`. All-Parasitics sums (decoupled + coupled + self C) are the net parasitics on each pin.

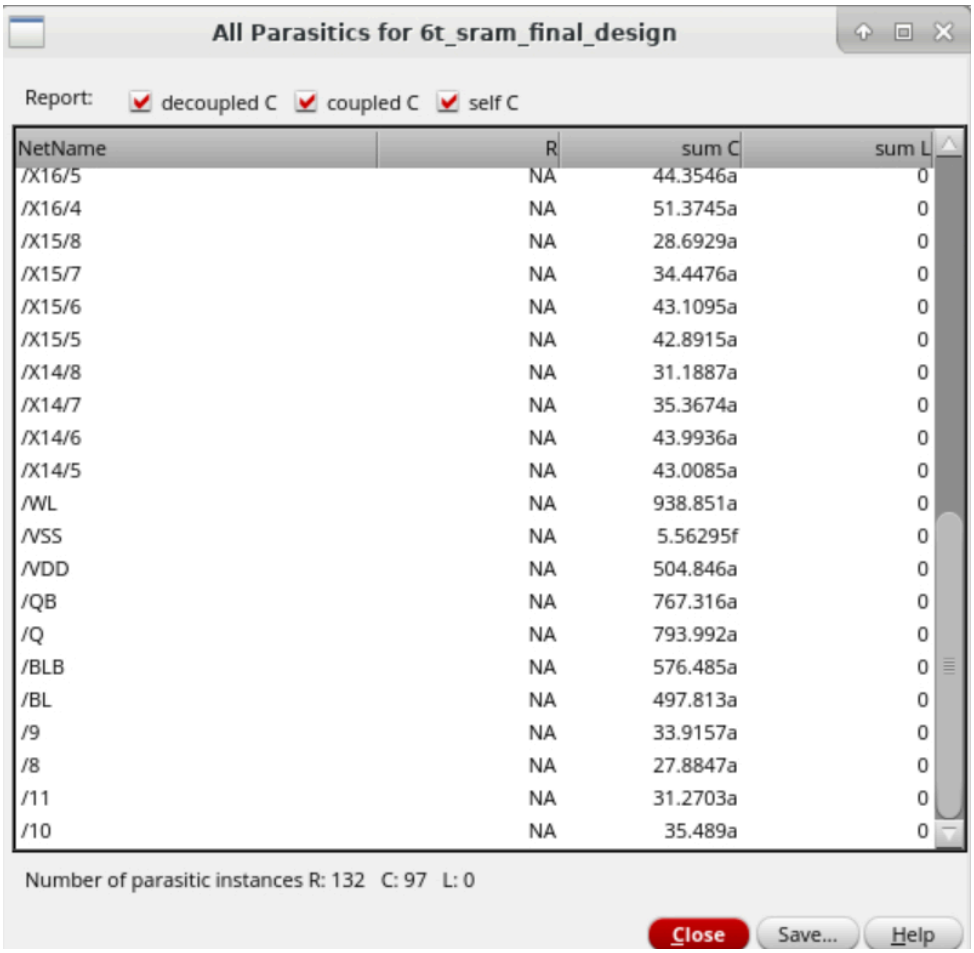


Figure 5: Quantus All Parasitics for `6t_sram_final_design`: $C_{BL} = 497.8$ aF, $C_{BLB} = 576.5$ aF, $C_{WL} = 938.9$ aF.

A column of 16 cells presents 16× the cell bitline net; a wordline of 4 cells presents 4× the cell wordline net. Adding 0.62 fF of M1/M2 array wire (the term used on the macro bench) gives the extracted column and row loads in Table 1. Those values are the C_{BL} and C_{WL} used for decoder effort, the read-pulse calculation, and the +20% recovery in Sections 2.3–4.

Table 1: Bitline and wordline capacitance from Part 1 Quantus, scaled to the 16×4 array.

Quantity	Part 1 cell	× cells	16×/4×	+ 0.62 fF wire
C_{BL}	497.8 aF	16	7.97 fF	**8.59** fF
C_{BLB}	576.5 aF	16	9.22 fF	**9.84** fF
C_{WL}	938.9 aF	4	3.76 fF	**4.38** fF

Column unit tests were closed at 3.5 fF and the decoder at 0.5 fF per wordline (the loads on those benches). Table 2 carries every measured delay to the extracted 8.59 fF and to +20% of extracted (10.3 fF) with the 3.5 → 4.2 fF slope for writes and C-proportional scaling for the bitline fall and precharge.

Table 2: Column delays at the test load, at +20% of the test load, and at the extracted array C_{BL}.

Quantity	3.5 fF	4.2 fF	8.59 fF	10.3 fF
Write, nfin 8 (ps)	96 meas.	102 meas.	140	154
t_{blb} to V_M (ps)	83 meas.	100	204	245
$t_{BL} = C\Delta V/I_{\text{cell}}$ (ps)	59	71	146	175
Precharge restore (ps)	124	149	304	364

Area (Part 1 convention: $D = 28$, $W_{\text{fin}} = 0.048$ μm, $L = 18$ nm, 0.0242 μm^2 per fin):

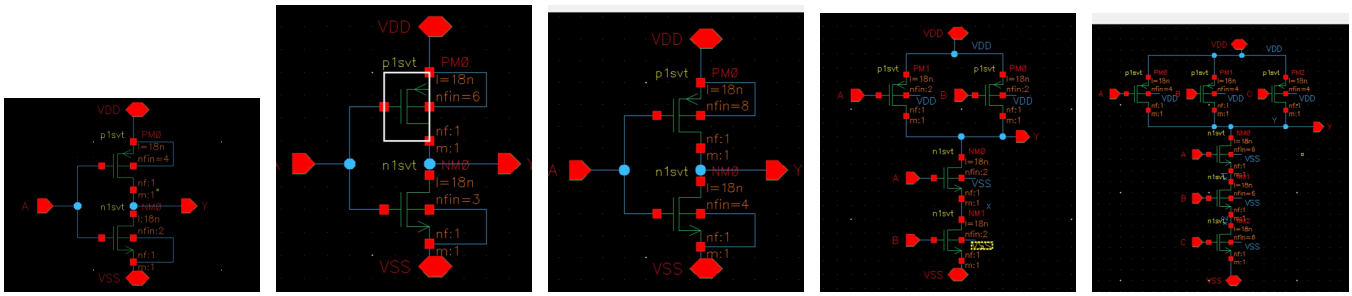
Block	Instances	Fins	Area, μm^2
Array, 2.88 × 8.64 μm (`6t_sram_64b`)	64	–	24.9
Row decoder	1	520	12.6
Controller (23 `inv_u`, 9 latches, 4 `inv_2x`, 3 `nand2_u`)	1	426	10.3
Write drivers / precharge / read paths / latches	4 each	144 / 24 / 16 / 96	3.5 / 0.58 / 0.39 / 2.3
Periphery = $D\sum W_i L_i$		1226	29.7
Macro total			**54.6**

The array term is the tiled PR-boundary area. The periphery is $D\sum W_i L_i$. Testbench buffers are excluded.

2 Row decoder and wordline timing (2.2)

2.1 Gates and sizes

Every gate is built from `inv_u` (PMOS 4 fins, NMOS 2 fins, $L = 18$ nm). `nand3_u` keeps that pull-down: NMOS 6 fins, PMOS 4. `nand2_u` is 2 fins on every device (half of `inv_u`; logical effort unchanged). `inv_2x` is 8/4; `inv_drv` is 6/3.



(a) `inv_u` **(b)** `inv_drv` **(c)** `inv_2x` **(d)** `nand2_u` **(e)** `nand3_u`

Figure 6: Decoder gates, fins per device, $L = 18$ nm.

2.2 Chosen implementation: 2-2 predecode

PA$\langle j\rangle$ = WL_EN·$(A_{1:0} = j)$ from `nand3_u`+`inv_2x`; PB$\langle m\rangle$ = $(A_{3:2} = m)$ from `nand2_u`+`inv_2x`; WL$\langle 4m + j\rangle$ = PA$\langle j\rangle$·PB$\langle m\rangle$ from `nand2_u`+`inv_drv`. Each literal reaches two predecode gates and each predecode output drives four rows, so $B = 8$ as the handout requires.

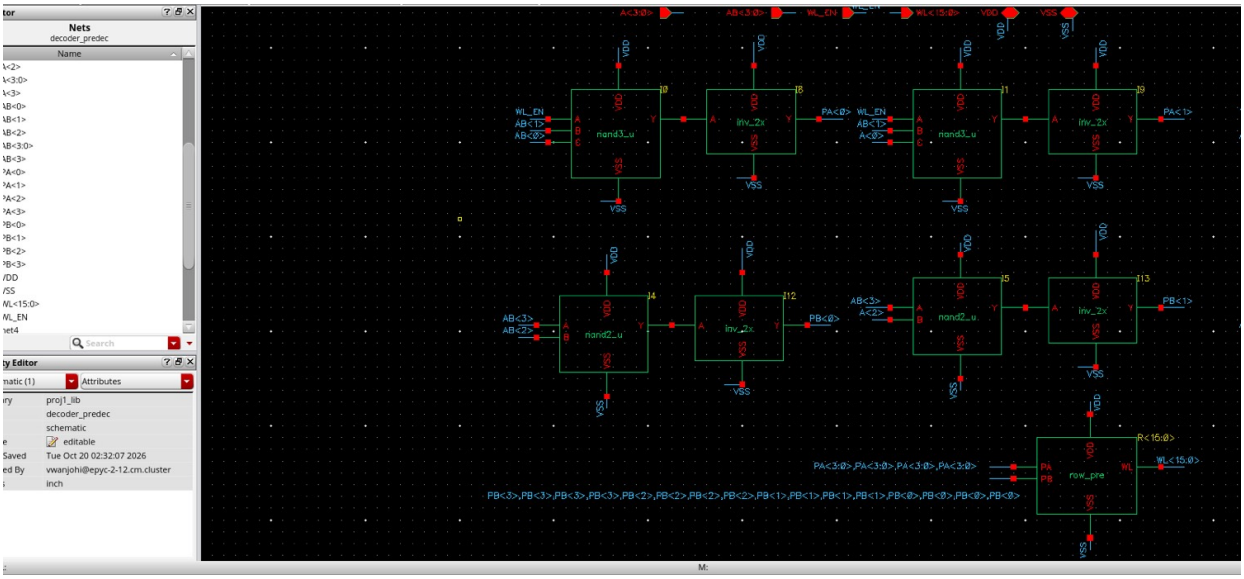


Figure 7: `decoder_predec`: PA (top), PB (middle) and sixteen `row_pre` instances.

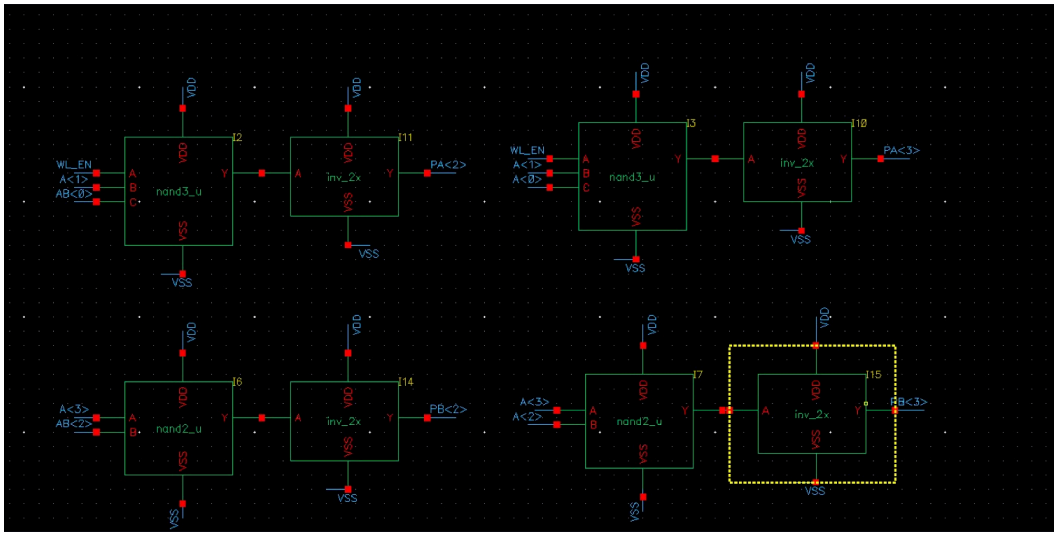


Figure 8: One PA NAND3, one PB NAND2 and one row NAND2+driver, as drawn.

2.3 Logical effort from an address input (2.2a)

FO4 of `inv_u` is 52.6 ps ($t_{pHL} = 54.7$ ps, $t_{pLH} = 50.6$ ps), so $\tau = 10.5$ ps. Gate capacitance ≈ 20 aF/fin. $p_{\mathrm{INV}} = 1$, $p_{\mathrm{NAND2}} = 2$, $p_{\mathrm{NAND3}} = 3$, $p_{\mathrm{NAND4}} = 4$. Electrical effort uses $C_{WL} = 4.38$ fF (Table 1) and, for the decoder bench, 0.5 fF.

The path from ADDR to WL is four stages. A_0 and A_1 (and WL_EN) enter `nand3_u`: NAND3–INV–NAND2–INV, $G = \frac{5}{3} \cdot 1 \cdot \frac{4}{3} \cdot 1 = \frac{20}{9} = 2.22$, $C_{in} = 10$ fins $= 200$ aF. A_2 and A_3 enter `nand2_u`: NAND2–INV–NAND2–INV, $G = \frac{4}{3} \cdot 1 \cdot \frac{4}{3} \cdot 1 = \frac{16}{9} = 1.78$, $C_{in} = 4$ fins $= 80$ aF. In both cases each literal fans out to two predecode gates and each predecode output to four rows, $B = 8$.

Optimal stage count $\hat{N} = \log_4 F$ with $F = GBH$. Four stages is the right count for the NAND3 path at the extracted C_{WL} ($\hat{N} = 4.3$). The NAND2 address path has a smaller C_{in} and a larger H; $\hat{N} = 4.8$, so four stages is slightly short of the effort-4 target and that path is the slower of the two address bits.

Table 3: Logical effort, address input → WL, 2-2 predecode.

Path	C_{WL}	H	F	$\hat{N}$	f	D
$A_{1:0}$ / WL_EN	0.5 fF	2.5	44	2.7	2.6	$17.3\,\tau = 182$ ps
$A_{1:0}$ / WL_EN	4.38 fF	21.9	389	4.3	4.4	$24.8\,\tau = 260$ ps
$A_{3:2}$	0.5 fF	6.3	89	3.2	3.1	$18.3\,\tau = 192$ ps
$A_{3:2}$	4.38 fF	54.8	779	4.8	5.3	$27.1\,\tau = 285$ ps

$N = 4$ is kept: adding a fifth inverter would cost a stage of parasitic delay for a modest drop in f on the $A_{3:2}$ path, and the measured WL_EN→WL delay is already 97 ps.

2.4 Second implementation: flat NAND4 (2.2b)

The alternative is one 4-input NAND of the address literals plus `inv_drv` on every row, with WL_EN as a fifth stacked NMOS (NAND5, $g = \frac{7}{3}$). Two stages. $G = \frac{7}{3}$, $B = 8$, $C_{in} \approx 8$ fins = 160 aF.

At $C_{WL} = 0.5$ fF: $H = 3.1$, $F = 58$, $f = 7.6$, $P = 6$, $D = 21.2\,\tau \approx 223$ ps. At $C_{WL} = 4.38$ fF: $H = 27.4$, $F = 511$, $f = 22.6$, $D = 51\,\tau \approx 540$ ps.

The 2-2 path keeps stage effort near 4 once C_{WL} is the extracted net; the flat path is effort-starved at two stages. Only A was built. Applying the measured/predicted ratio of A (97/182 = 0.53) to B gives the schematic delays in the last column of Table 4.

Table 4: Predicted WL_EN→WL delay versus the measured 2-2 predecode. The last column scales B by A's 97/182 ratio.

Implementation	Predict, 0.5 fF	Predict, 4.38 fF	Measured, 0.5 fF	Scaled, 0.5 fF
2-2 predecode (A)	182 ps	260 ps	**97 ps**	97 ps
Flat NAND4+driver (B)	223 ps	540 ps	—	119 ps

A is faster at both loads. The 0.5 fF prediction is conservative (1.9× the measured 97 ps); the extracted C_{WL} raises the prediction to 260 ps, still inside the 500 ps clock high at 1 ns and inside the 345 ps high at $T_{CLK,\min}$.

2.5 Measured decoder delay

`tb_dec_pre`, 0.5 fF per wordline, WL_EN 300 ps with 20 ps edges, static address. WL_EN→WL is 97.4 ps rising and 101 ps falling at every address (WL_EN always enters the same `nand3_u` pin). Peak of the summed wordlines is 824 mV at all 16 addresses: exactly one row high (< 1 V) and it reaches high (> 0.72 V). `wl_idx` (row index weighted by wordline peak over 0.824 V) returns the selected row:

Address	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
`wl_idx`	0.06	1.06	2.06	3.06	4.03	5.04	6.04	7.04	8.01	9.01	10.01	11.01	11.98	12.98	13.99	14.99

Each value is distinct and within 0.07 of its row. Offsets are ≈ 2 mV coupling on rows sharing PA⟨0⟩.

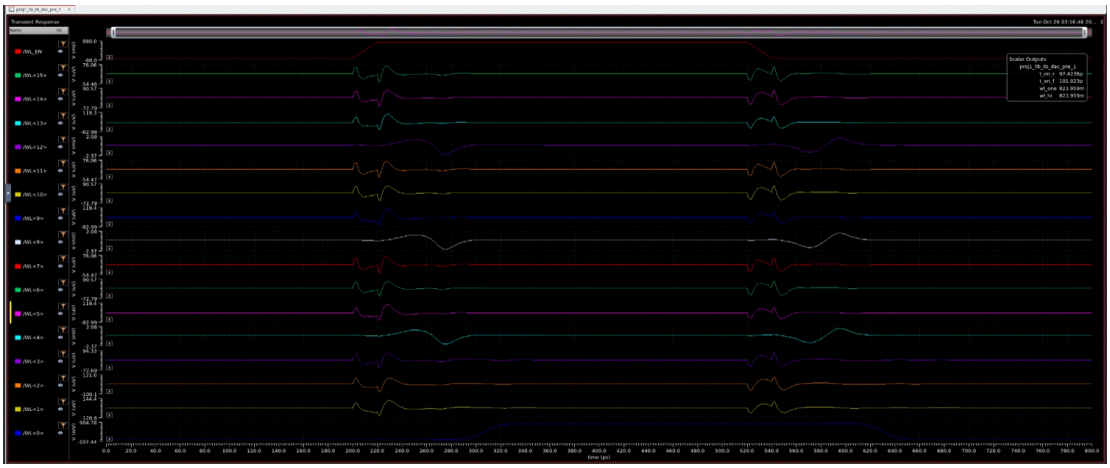


Figure 9: Address 0: only WL⟨0⟩ rises; unselected rows stay within 2 mV.

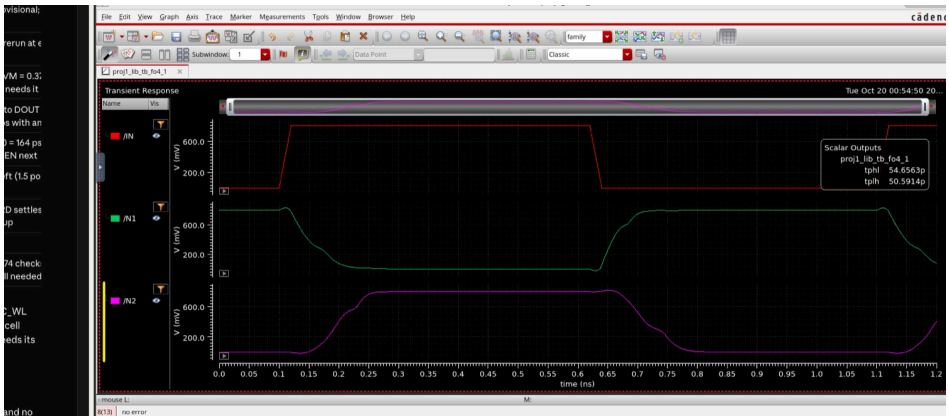


Figure 10: `tb_fo4`: IN, N1, N2 of an `inv_u` chain. FO4 = 53 ps.

2.6 Wordline pulse width (2.2c)

The read needs the bitline to fall $\Delta V = V_{DD} - V_M = 0.424$ V through the cell current $I_{\text{cell}} = 25\ \mu$A (Part 1, 0.8 V):

$$t_{BL} = \frac{C_{BL}\,\Delta V}{I_{\text{cell}}}. \tag{1}$$

At the column-test load $C_{BL} = 3.5$ fF this is 59 ps. The measured BLB time to V_M is 83 ps; the extra 24 ps is the drop in I_{cell} as V_{DS} of the pass-gate falls. At the extracted $C_{BL} = 8.59$ fF the same formula gives 146 ps, and C-scaling of the measured 83 ps gives 204 ps.

Quantity	Value	Source
t_{BL}, 3.5 fF, $C\Delta V/I_{\text{cell}}$	59 ps	$I_{\text{cell}} = 25\ \mu$A, $\Delta V = 0.424$ V
t_{blb} measured	83 ps	read-column test, 3.5 fF
t_{BL}, extracted 8.59 fF	146 ps	same I_{cell}
Read, minimum WL at 50%	110 ps	83 ps plus edge margin, 3.5 fF
Write, minimum WL at 50%	100–120 ps	write-driver sweep, 3.5 and 4.2 fF
Write at extracted 8.59 fF	140 ps	Table 2
Design WL pulse, 50% width	**140 ps**	set by the write at the column-test load

The write sets the pulse at 3.5 fF. At the extracted column the I_{cell} bound is 146 ps and the scaled write is 140 ps, so the 140 ps pulse is tight. Recovery is two extra inverters on DLY1 ($\approx$ 20 ps on WL_EN) or a 50 ps stretch of the clock high (Section 4).

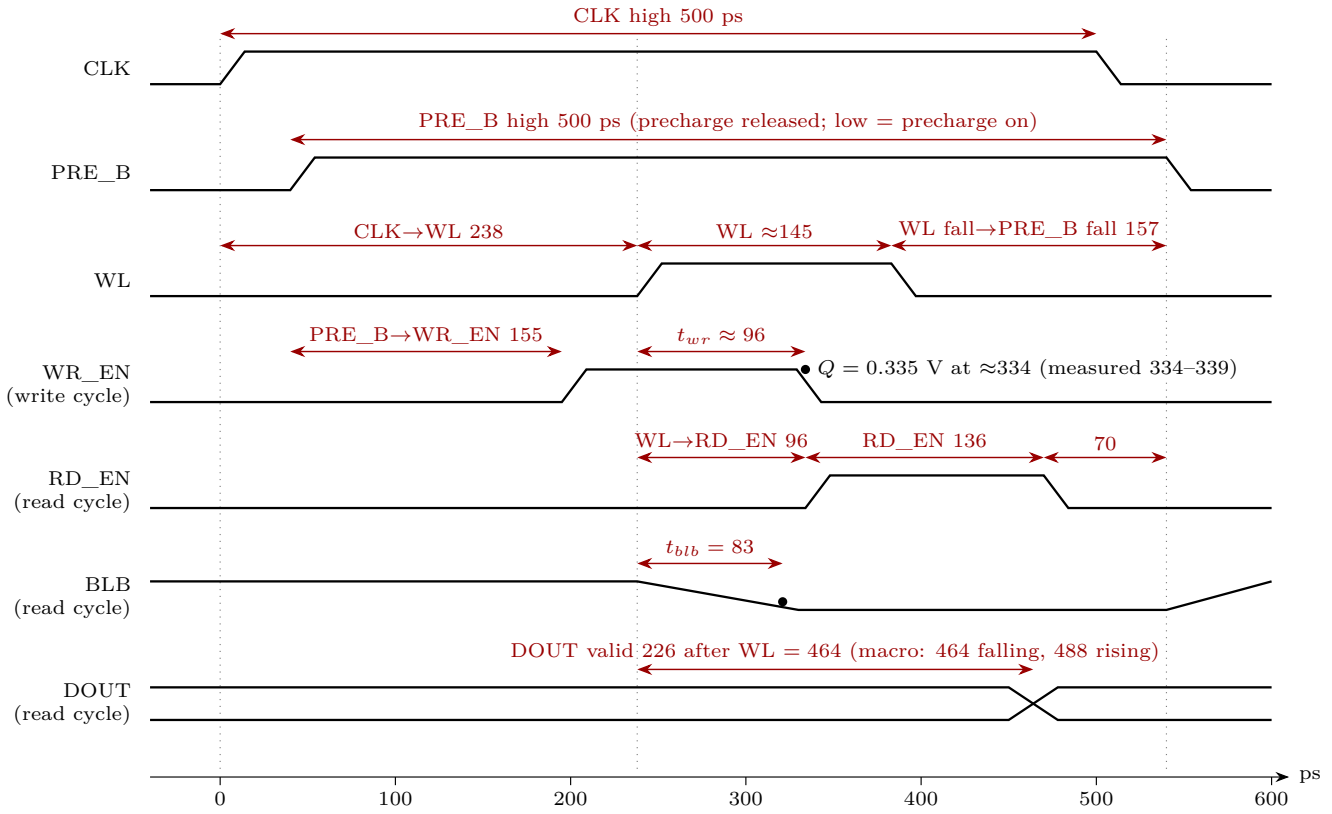


Figure 11: Annotated timing, $t_{clk} = 1$ ns, times in ps from the rising CLK edge. Measured: CLK→WL 238, WL pulse ≈ 145, WL→RD_EN 96, RD_EN width 136, WL and RD_EN fall 157 and 70 ps before PRE_B, PRE_B rises 155 ps before WR_EN, $t_{blb} = 83$, write ≈ 96 after WL, DOUT 226 after WL.

3 Column circuits (2.3)

Unit tests use $C_{BL} = 3.5$ fF. Extracted and +20% extracted loads are in Table 2.

3.1 Precharge

Three PMOS, 2 fins each: one per bitline to VDD and one equalizer, gated by PRE_B. PRE_B follows CLK, so precharge is on while CLK is low and off while CLK is high. At $T_{CLK,\min} = 0.69$ ns the low phase is 345 ps, which is the recovery window. The 2-fin `p1svt` is taken at $I_{\text{on,p}} = 12$ μA (≈ 0.5× the 25 μA n-pass of the cell). Restore of $\Delta V = V_{DD} - V_M$ is

$$t_{\text{pre}} = \frac{C_{BL}(V_{DD} - V_M)}{I_{\text{on,p}}}. \tag{2}$$

That is 124 ps at 3.5 fF, 304 ps at 8.59 fF, and 364 ps at 10.3 fF. The 3.5 fF and extracted cases finish inside the 345 ps low. At +20% of extracted, t_{pre} overruns the half-period by 19 ps; recovery is a 40 ps stretch of T_{CLK} (0.73 ns) or a 45% duty so the low phase is 380 ps. In every passing column test the bitlines are already at VDD when WL rises.

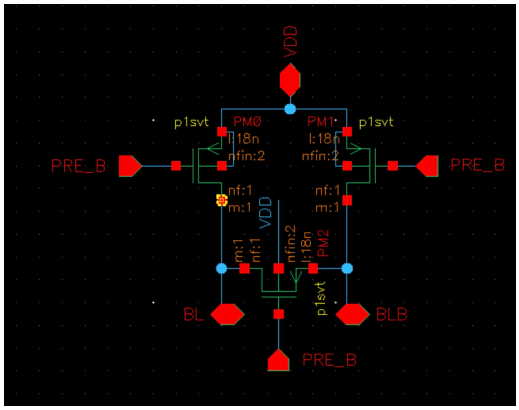


Figure 12: `precharge`: three 2-fin PMOS, gate PRE_B.

3.2 Write driver

Four NMOS pull-downs plus a 2+2-fin inverter. A small driver loses the race against the undriven bitline: the cell discharges BLB through its pass-gate, pins $\overline{Q}$ low and blocks its own feedback.

Table 5: Driver sizing ($V_{DD} = 0.8$ V, $C_{BL} = 3.5$ fF, $pw_w1 = 0.3$ ns).

nfin	Write 0	t_{wr0} (ps)	t_{wr1} (ps)	Lowest BLB (mV)
2	fail, Q stalls at 0.66 V	–	–	∼0
4	fail, Q stalls at 0.38 V	–	–	∼60
5	pass	153	154	∼190
6	pass	121	121	∼260
8	pass	96	96	∼345

WL at 50%	$C_{BL} = 3.5$ fF	4.2 fF (+20% of 3.5 fF)
∼60, 80, 100 ps	fail	fail
∼120, 140, 170 ps	pass, 96 ps	pass, 102 ps

nfin 5 is the smallest that writes. nfin 8 writes in 96 ps (102 ps at +20% of 3.5 fF) and is the design point: 0.77 μm^2 over nfin 6 (3% of the array) buys 25 ps, and $T_{CLK,\min}$ is set by the read, not by this 6 ps. The 3.5 → 4.2 fF slope is 8.6 ps/fF, so the extracted 8.59 fF writes in 140 ps and +20% extracted (10.3 fF) in 154 ps. Minimum write pulse 100–120 ps at 50% at the test loads; 160 ps at 10.3 fF.

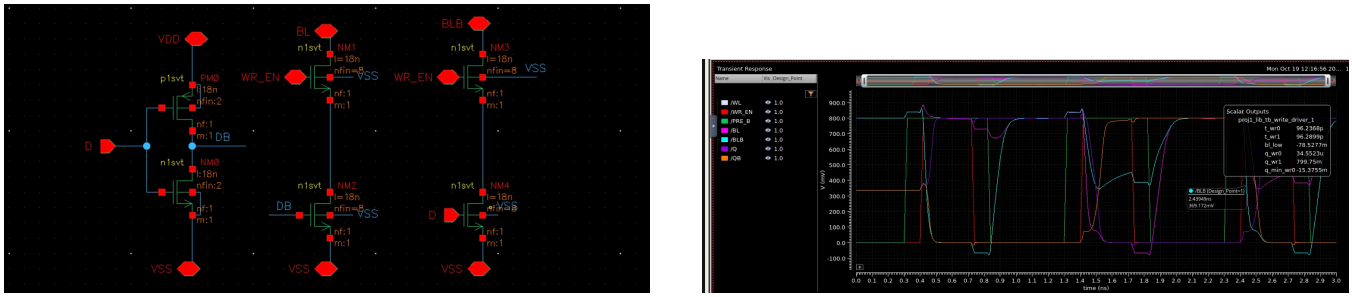


Figure 13: Write driver at nfin 8, and write-0 / write-1 ($q_{wr0} = 35$ μV, $q_{wr1} = 799.8$ mV, $t_{wr0} = 96$ ps).

3.3 Read path

Full-swing inverter on BLB, no sense amplifier. Chosen 2:2 (PMOS 2, NMOS 2), $V_M = 376$ mV, so BLB must fall 0.424 V to flip RD. The 140 ps WL pulse gives 57 ps of slack over the 83 ps bitline time at 3.5 fF. A 4:2 inverter would switch at 0.415 V, save ≈ 7 ps and add 8 fins (0.19 μm^2); it is off the critical path because the write sets the pulse.

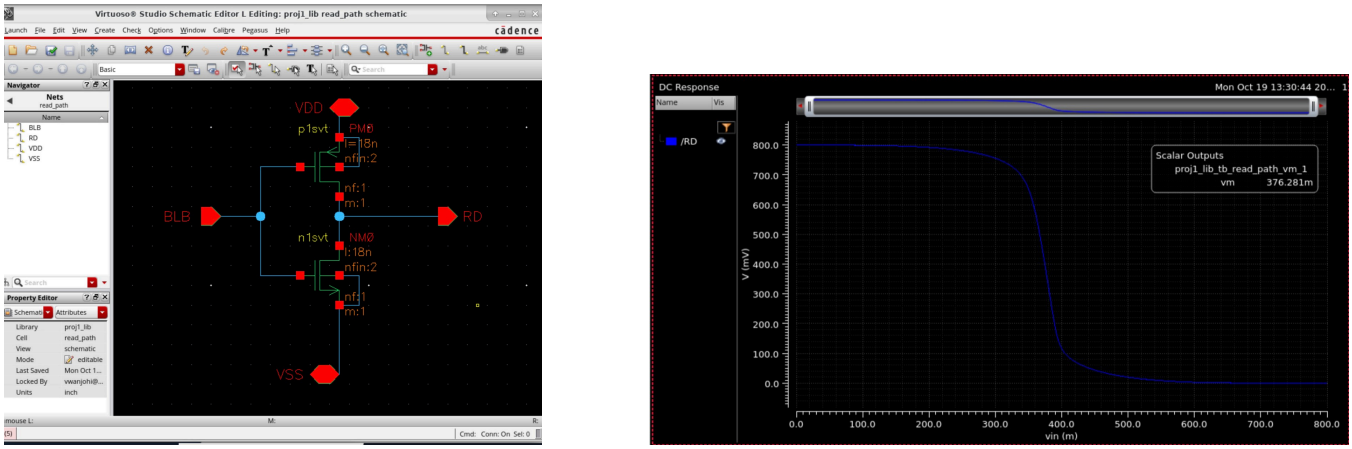


Figure 14: `read_path` (2:2) and its transfer curve, $V_M = 376$ mV.

3.4 Output latch

All devices 2 fins. Enable-pulse sweep, $c_{dout} = 1$ fF; hold-1 > 0.72 V, hold-0 < 0.08 V. 10 ps fails (never captures a 1). 20 ps and wider pass. Design width 120 ps (column test).

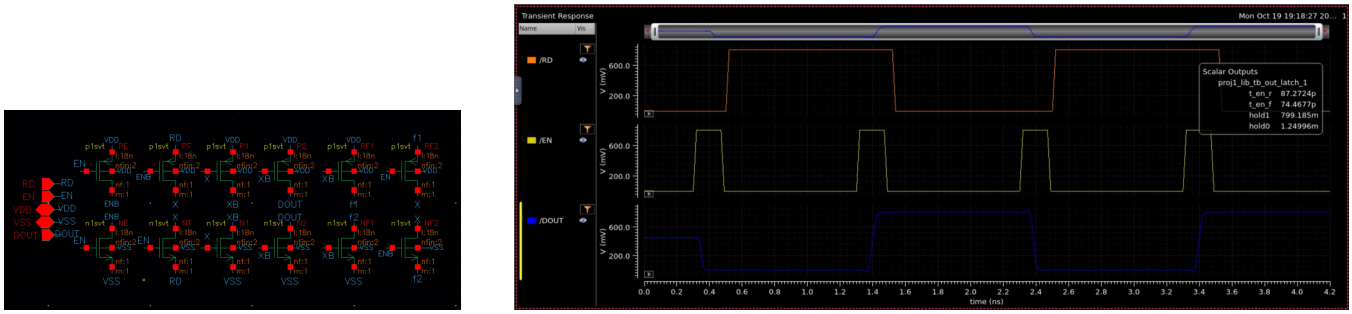


Figure 15: `out_latch` and unit test at 150 ps: DOUT writes only while EN is high.

3.5 Read column

Cell, write driver, precharge, read path and latch together: write 1, read, write 0, read, write 1, read. Sweep of latch delay `rd_dly` and width `pw_rden` (24 points). All checks pass except the third read at `pw_rden`= 40 p.

`rd_dly`	40p		80p	120p	160p	t_{acc0}
1.48n	fail		247	**226**	226	165
1.52n	fail		235	235	235	205
1.56n–1.68n	fail	+40 ps per 40 ps of delay				+40 ps

Chosen: `rd_dly`= 1.48 n, `pw_rden`= 120 p. BLB hits 0.376 V 83 ps after WL; DOUT valid 165 ps (read 0) and 226 ps (read 1). Cell undisturbed: $q_{rd1} = 799.7$ mV, $q_{rd0} = 39$ μV.



Figure 16: Cycle-6 read of a 1, $C_{BL} = 3.5$ fF: BLB crosses 0.376 V at 83 ps; DOUT valid 226 ps after WL.

4 Controller (2.4)

Inputs CLK, WE, ADDR, DIN. Outputs A, AB, D, PRE_B, WL_EN, WR_EN, RD_EN. Input latches (`out_latch`, enable CLKBL from its own `inv_2x`) hold ADDR, DIN and WE while CLK is high. PRE_B = `inv_2x`(CLKB) follows CLK; the precharge PMOS conduct when PRE_B is low, so precharge is on while CLK is low. WL_EN = CLKBUF AND (CLKBUF through 7 `inv_u`): a one-shot on each rising edge. WR_EN = WL_EN AND WE. RD_EN = (WL_EN through 8 `inv_u`) AND $\overline{\text{WE}}$.

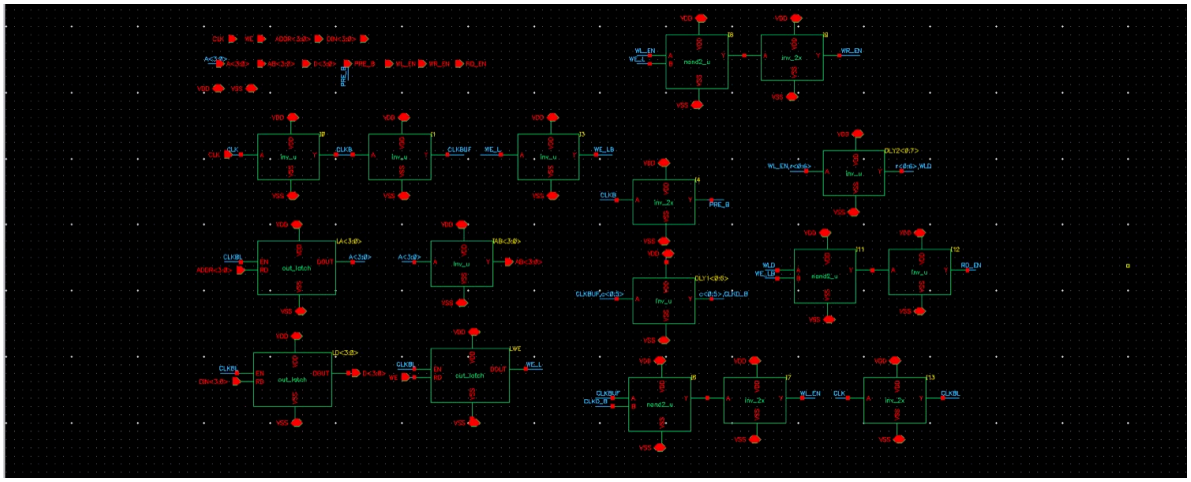


Figure 17: `controller`: input latches, WL_EN one-shot (DLY1), WR_EN and RD_EN (DLY2).

Table 6: Controller timing at 0.4 V crossings, before and after moving the latch enable onto its own inverter.

Output	Before	After	
CLK → WL	356 ps	**238 ps**	
RD_EN fall before PRE_B fall	30 ps	**70 ps**	thinnest margin
WL fall before PRE_B fall	118 ps	157 ps	
PRE_B rise before WR_EN rise	178 ps	155 ps	
WL → RD_EN	96 ps	96 ps	column needs ≥ 80
RD_EN width	136 ps	136 ps	column needs ≥ 100
WL_EN width	134 ps	134 ps	write needs 120–140

No spurious pulses: RD_EN peaks at 3.3 mV in a write cycle, WR_EN at 0.41 mV in a read cycle. Margins after the CLKBL change: WL_EN width +14 ps over 120 ps; RD_EN lead +16 ps over 80 ps; RD_EN width +36 ps over 100 ps; RD_EN before precharge +70 ps; WL before precharge +157 ps; PRE_B before WR_EN +155 ps.

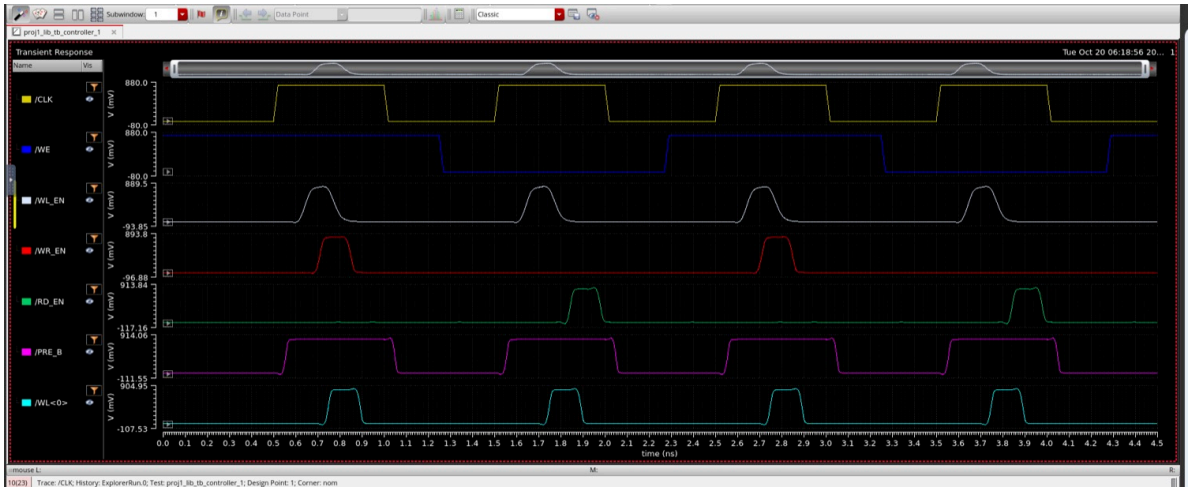


Figure 18: Controller transient: WR_EN in write cycles only, RD_EN in read cycles only; WL ends before PRE_B falls.

+20% of extracted C_{BL}. Extracted $C_{BL} = 8.59$ fF; +20% is 10.3 fF. From Table 2: write 154 ps, t_{blb} 245 ps, t_{pre} 364 ps. The 140 ps WL pulse and the 345 ps CLK-low window are both short of those three numbers. Recovery, without resizing the cell:

- DLY1: two extra `inv_u` (≈ 20 ps) on WL_EN, then two more if the scaled t_{blb} is the bound, for a ≈ 180 ps pulse.
- DLY2: drop two inverters (≈ 20 ps) so RD_EN still opens after the bitline crosses V_M.
- Clock: stretch T_{CLK} to 0.73 ns, or keep a 45% duty so CLK-low is 380 ps, covering the 364 ps restore.

At the column-test +20% (4.2 fF) none of this is needed: write +6 ps, t_{blb} 83 → 100 ps, both inside 140 ps. The 70 ps RD_EN-to-precharge margin is the one that shrinks if the read stretches. The 70 ps margin does not set $T_{CLK,\text{min}}$: it would have vanished near 0.86 ns if the half-period shrank around a fixed pulse, yet the macro passes to 0.69 ns.

5 Verification (2.5)

`tb_sram_macro`: CLK, WE and ADDR through two `inv_u` each; 1 fF on DOUT. Address is a binary counter, one step per cycle. WE high for 16 cycles (write all words), low for 16 (read all), repeating. DIN $= \{X, \bar{X}, X, \bar{X}\}$ with $X = a_0 \oplus ph$, so each word is 0101 or 1010 alternating by row; ph flips at $32.25\,t_{clk}$ and complements every word. `err1`/`err2` are the peak DOUT error in the checkerboard and complement read windows against 0.1 V; `err3` is test (iii).

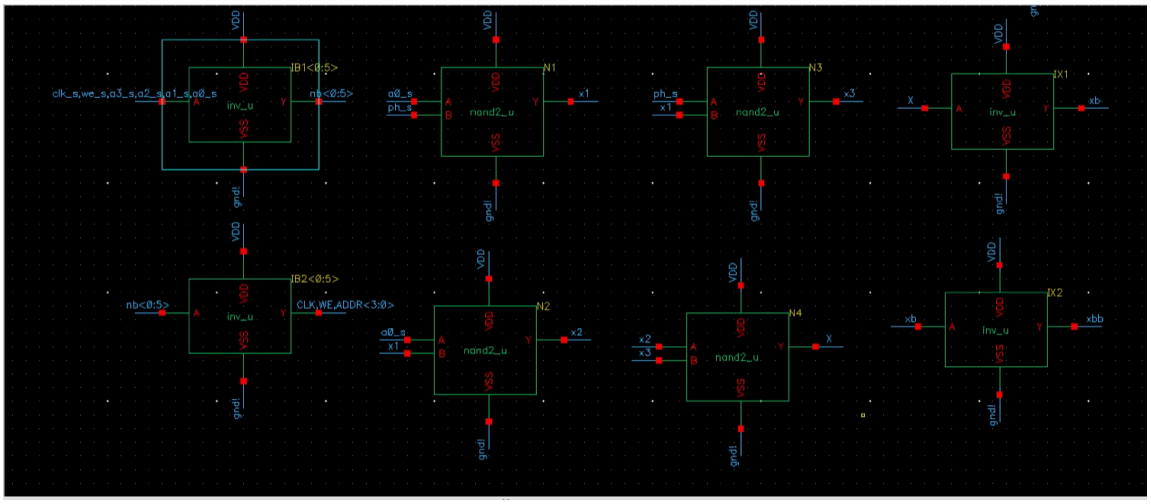


Figure 19: Input buffers and the NAND XOR that builds DIN.

Table 7: Pass/fail at $t_{clk} = 1$ ns.

Test	Error	
(i) checkerboard write, read all 16	`err1` $= 174\ \mu$V	**pass**
(ii) complement overwrite, read all 16	`err2` $= 174\ \mu$V	**pass**
(iii) row 15, read immediately after a write of the opposite value	`err3` $= 174\ \mu$V	**pass**

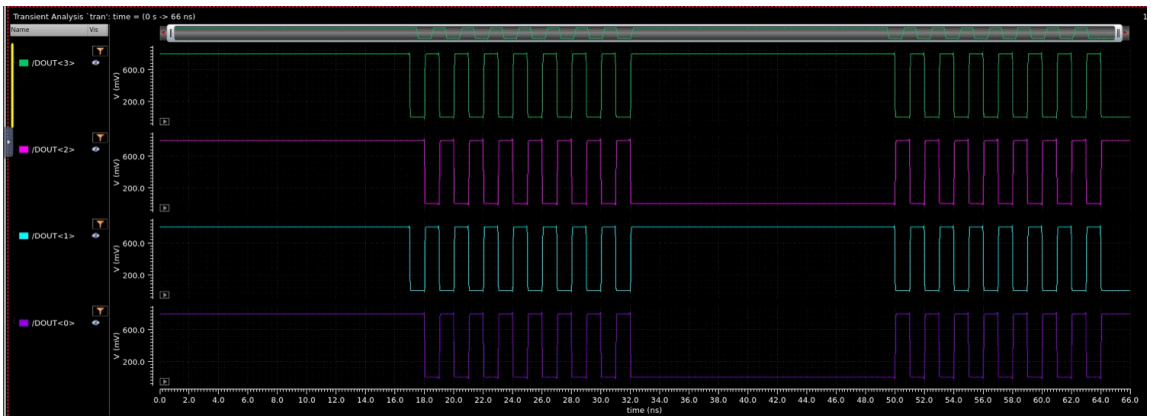


Figure 20: DOUT⟨3:0⟩: bits 3,1 opposite bits 2,0. DOUT holds while WE is high.

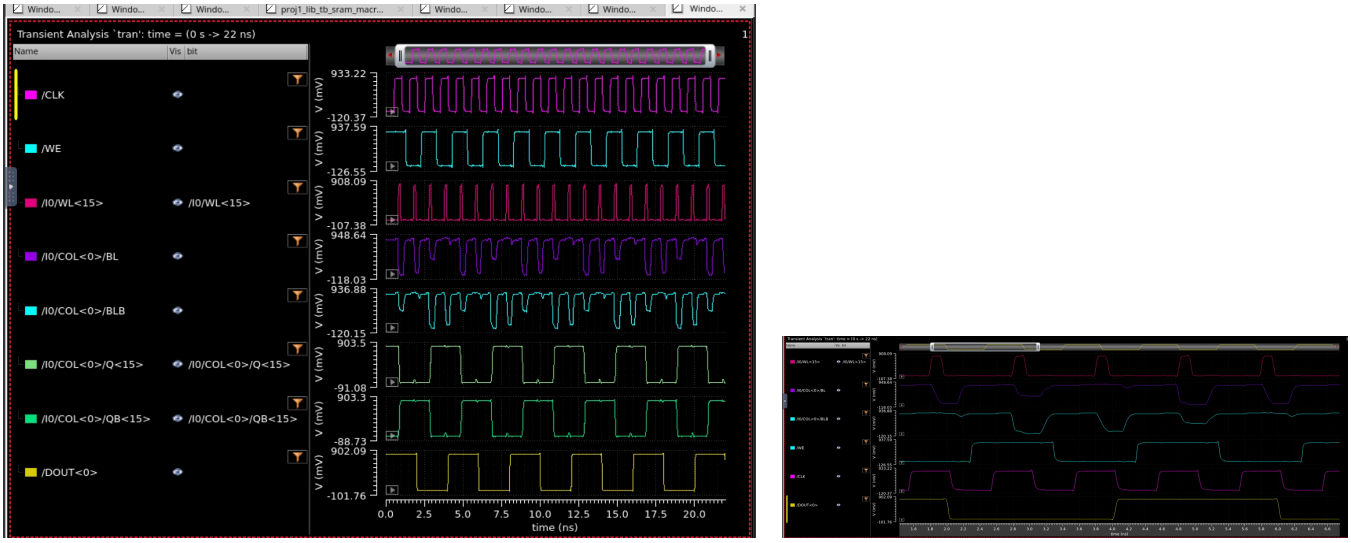


Figure 21: Test (iii), row 15. WE alternates every cycle; each read returns the value just written. DOUT changes ≈ 0.5 ns after the read edge.

Write time (later of Q and $\overline{Q}$ crossing 0.335 V, from the CLK edge, row 15): 339 ps (write 1), 334 ps (write 0). That is the controller's 238 ps to WL plus the column's 96 ps. **Read access** (CLK to DOUT⟨1⟩, 0.4 V): 464 ps falling, 484 ps rising at 1 ns; 488 ps rising at $T_{CLK,\min}$.

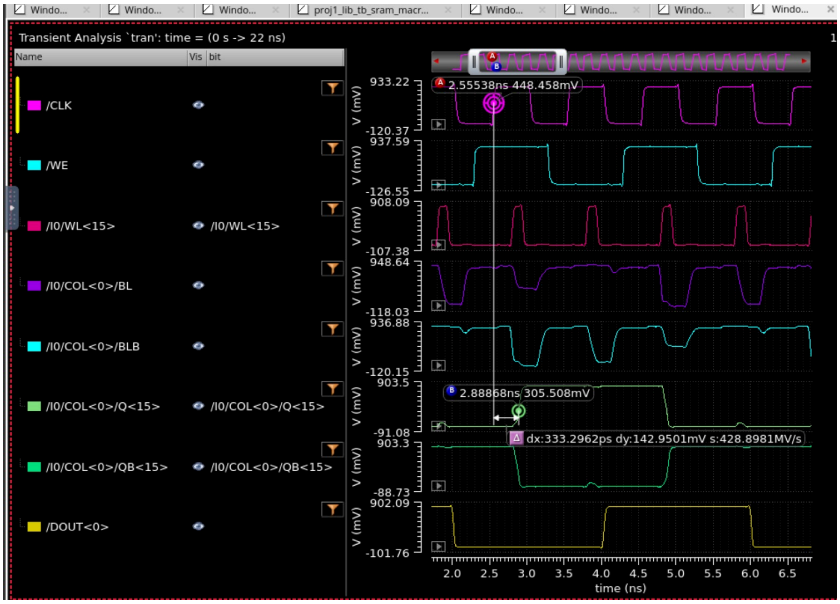


Figure 22: Write time on row 15: CLK edge to $Q\langle 15\rangle$, 333 ps on the cursors (339 ps at the 0.335 V crossing).

Table 8: Clock-period sweep. Power is the macro VDD-pin current ×0.8 V.

t_{clk}	err1	err2		$t_{acc,f}$	$t_{acc,r}$	write	read
1 ns – 750 ps	174 μV – 0.50 mV	same	pass	464	484–486	–	–
700 ps	5.10 mV	5.12 mV	pass	464	486	123 μW	113 μW
690 ps	23.3 mV	22.4 mV	**pass**	464	**488**	124 μW	115 μW
680 ps	452 mV	441 mV	fail	464	497	126 μW	118 μW
≤ 670 ps	1.28 V	1.28 V	fail	464	–	128–143 μW	115–123 μW

`err3` tracks the same cliff: ≈ 30 mV at 690 ps, ≈ 0.6 V at 680 ps, 1.28 V below that. Write times stay at 339/334 ps. $T_{CLK,\min} = 0.69$ ns (10 ps resolution).

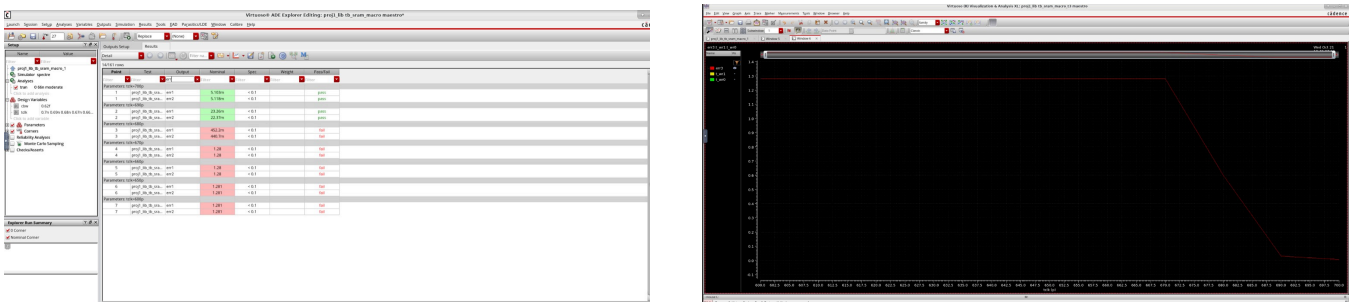


Figure 23: Error versus clock period: checkerboard/complement (left) and worst-row (right). Fail below 0.69 ns.

6 Metrics and figure of merit

$V_{DD} = 0.8$ V, nominal, 27°C, $D = 28$. Write power is the macro VDD-pin current ×0.8 V with WE high every cycle and all four columns writing (the checkerboard pass at $T_{CLK,\min}$). That is the same four-column write current as an all-0 then all-1 sweep: every cycle the write drivers fight the precharged bitlines. Standby with the clock running still fires WL_EN on every edge (no chip-select); true standby is the clock stopped.

Table 9: Macro metrics at $T_{CLK,\min}$.

Metric	Value
$T_{CLK,\min}$	**0.69** ns (690 ps passes (i)–(iii); 680 ps fails)
Worst-case read access, CLK→DOUT	488 ps (DOUT⟨1⟩ rising at 690 ps)
Worst-case write time, CLK→ $Q/\overline{Q}$ past 0.335 V	339 ps (row 15)
Array write power at $T_{CLK,\min}$	124 μW (WE high, four columns writing)
Checkerboard read power	115 μW
Standby, CLK running (WL_EN still fires)	112 μW
Standby, CLK stopped	3.99 μW
Area: array 2.88 × 8.64 μm + periphery	$24.9 + 29.7 = \mathbf{54.6}$ μm^2
FOM = Area · P_{write} · $T_{CLK,\min}$	$54.6 \times 124 \times 0.69 \approx \mathbf{4.7 \times 10^3}$ μm^2 · μW·ns

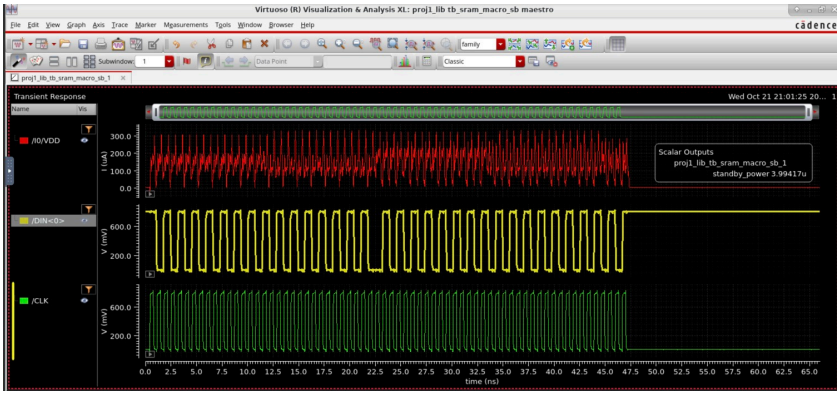


Figure 24: Standby: after the clock stops (≈ 47 ns) the VDD current falls to leakage, 3.99 μW.

Trade-offs

- **Write driver.** nfin 5 is the smallest that writes (153 ps). nfin 8 is 96 ps for 0.77 μm^2 over nfin 6. Kept, because T_{CLK} is set by the read.
- **Read inverter 2:2.** Saves 8 fins versus 4:2 and costs ≈ 7 ps, off the critical path.
- **2-2 predecode versus flat NAND4.** Predicted 182 ps versus 223 ps at 0.5 fF, and 260 ps versus 540 ps at the extracted C_{WL}. Measured 2-2 delay 97 ps. Four stages match $\hat{N} = 4.3$ on the $A_{1:0}$ path.

- **Latch enable.** Opening RD_EN at 1.48 ns gives the shortest access (226 ps) with no glitch; 120 ps is the shortest pulse that still captures.
- **Controller CLKBL.** A dedicated latch-enable inverter cut CLK→WL from 356 to 238 ps and doubled the thinnest margin (30 → 70 ps) for 12 fins. The periphery is 54% of the macro, mostly the nine input latches.
- **Clocked standby.** There is no chip-select, so a running clock always accesses the array: 112 μW versus 4 μW with the clock stopped.
- **Bitline load.** Sizes closed at 3.5 fF. Extracted $C_{BL} = 8.59$ fF. Write 96 → 140 ps; $t_{BL} = 146$ ps from I_{cell}. At +20% extracted (10.3 fF) recover by lengthening DLY1, shortening DLY2, and stretching the clock high or the CLK-low duty—not by resizing the cell.